

Two-Way Tables & Inference for a Single Mean

IMS Chapters 18 & 19

Inference for two-way tables

Beyond two groups

Chapter 17 compared proportions across **two** groups. But what if a variable has **three or more** categories on either side?

*Researchers had 219 participants sell a used iPod (known to have frozen twice) to a buyer who asked one of three questions: a **general** question, a **positive-assumption** question (“it doesn’t have problems, does it?”), or a **negative-assumption** question (“what problems does it have?”). Did the question asked affect whether the seller disclosed the freezing issue?*

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With 3 categories on one side, “the difference in proportions” no longer has a single, obvious meaning — we need a new tool.

The data

Question	Disclosed	Hid	Total
General	2	71	73
Positive assumption	23	50	73
Negative assumption	36	37	73
Total	61	158	219

H_0 : the question asked and disclosure are **independent**. H_A : they are **not** independent.

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The disclosure rates look very different across rows (2.7%, 31.5%, 49.3%) — but is that a real effect, or could chance alone produce a spread this large?

Expected counts

If disclosure really didn't depend on the question, **every** row should show the same overall disclosure rate: $61/219 = 0.2785$.

i Expected count formula

$$\text{Expected count}_{\text{row } i, \text{col } j} = \frac{(\text{row } i \text{ total}) \times (\text{col } j \text{ total})}{\text{table total}}$$

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Observed was only 2 — a big gap from what independence would predict.

The chi-squared statistic

For every cell, compare observed to expected:

$$X^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

$$\text{Row 1, Col 1: } \frac{(2 - 20.33)^2}{20.33} = 16.53 \quad \dots \quad \text{Row 3, Col 2: } \frac{(37 - 52.67)^2}{52.67} = 4.66$$

Summing over all 6 cells: $X^2 = 40.13$.

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Bigger gaps between observed and expected $\rightarrow$ bigger $X^2 \rightarrow$ stronger evidence against independence.

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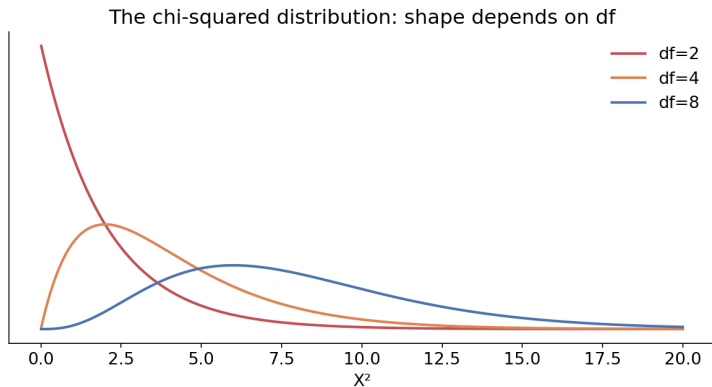
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After 1,000 shuffles: **not a single one** came anywhere close to $X^2 = 40.13$.

The simulated p-value is effectively **zero** — extremely strong evidence that the question asked really did change sellers' disclosure behavior. (Causal language is fine here — this was an experiment.)

The mathematical model: chi-squared distribution

χ^2 has its own named sampling distribution — the **chi-squared distribution** — indexed by degrees of freedom.



DoF

i Degrees of freedom for a two-way table

$$df = (R - 1) \times (C - 1)$$

For our 3×2 table: $df = (3 - 1)(2 - 1) = 2$.

Finding the p-value

Conditions: independent observations; every expected count ≥ 5 .

$$X^2 = 40.13, \quad df = 2$$

In R: `1 - pchisq(40.13, df = 2)  $\approx$  0.0000000019.`

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Since this is far below $\alpha = 0.05$, we **reject** H_0 — the question asked and disclosure are not independent. Note: a chi-squared test only tells us the variables are *related*, not which specific question drives the effect.

In code

```
ipod_table <- table(ask$question_class, ask$response)

# Mathematical model
chisq.test(ipod_table)

# Randomization equivalent: shuffle response, recompute  $X^2$ , repeat

chisq.test() reports the same  $X^2$ ,  $df$ , and p-value we just found by hand.
```

Inference for a single mean

From proportions to means

Everything so far this unit involved **categorical** outcomes. Now: what if the response is **numerical**?

You're considering buying a franchise of the used-car dealer Awesome Autos.

You sample 5 cars and want to estimate the true average selling price across all their cars.

Same three tools as always (randomization doesn't quite apply to a single mean the same way, but bootstrapping and mathematical models both do) — just now the statistic is $\bar{x}$ instead of $\hat{p}$.

Bootstrapping a mean

Our sample of 5 cars: mean $\bar{x} = \$17,140$, $s = \$7,170.29$.

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The resulting bootstrap distribution of $\bar{x}_{boot}$ values lets us build a **percentile confidence interval** — sort the bootstrapped means, take the 2.5th/97.5th percentile for a 95% CI — exactly as before.

Two ways to build the bootstrap CI

i Bootstrap percentile CI

Take the appropriate percentiles of the bootstrap distribution directly.

i Bootstrap SE CI

$$\text{point estimate} \pm 2 \times SE_{boot}$$

(using “2” as an approximation tied to the 68-95-99.7 rule)

For Awesome Autos: percentile CI $\approx$ (\$11,778, \$22,500); SE CI $\approx$ \$17,140 $\pm$ 2(\$2,891.87) = (\$11,356, \$22,924).

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Different methods, not identical numbers — but reassuringly close.

A caution about bootstrapping

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Bootstrapping is **not** a fix for having too small a sample.

It approximates the variability of $\bar{x}$ well when the original sample is large and representative — it does not manufacture information that isn't there. A bootstrap CI from 5 cars is still built on just 5 cars' worth of information.

Enter the t-distribution

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The catch, again: we essentially never know σ . Substituting s introduces extra uncertainty — enough that the normal distribution is the wrong shape to use. We need the **t-distribution** (heavier tails, indexed by $df = n - 1$) instead.

Checking conditions for $\bar{x}$

- ▶ **Independence:** simple random sample (or a legitimately random process).
- ▶ **Normality:** for small n , no clear outliers in the data; for large n , no *particularly extreme* outliers (slight skew is fine at $n \approx 15$, moderate skew at $n \approx 30$, strong skew at $n \approx 60$).

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These conditions are a bit fuzzier than the success-failure condition for proportions — this takes practice to judge well.

A t-interval: dolphin mercury content

Sample of $n = 19$ Risso's dolphins: mean mercury content $\bar{x} = 4.4$, $s = 2.3$ ($\mu\text{g/wet gram}$).

$$SE = \frac{s}{\sqrt{n}} = \frac{2.3}{\sqrt{19}} \approx 0.528, \quad df = 19 - 1 = 18$$

For a 95% CI, $t_{18}^* = 2.10$ (found via `qt(0.975, df=18)`).

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We are 95% confident the true average mercury content is between 3.29 and 5.51 $\mu\text{g}/\text{wet gram}$ — considered extremely high.

The one-sample t-test

i T statistic for a single mean

$$T = \frac{\bar{x} - \text{null value}}{s/\sqrt{n}}, \quad df = n - 1$$

Mechanically identical to the Z-formula from before — same “estimate minus null, over SE” logic — just paired with the t-distribution instead of the normal, and called T instead of Z as a reminder of that.

Worked example: are runners getting slower?

2006 Cherry Blossom 10-mile race average: 93.29 minutes. A 2017 sample of 100 runners: $\bar{x} = 98.78$, $s = 16.59$.

$H_0 : \mu = 93.29$ vs. $H_A : \mu \neq 93.29$.

$$SE = \frac{16.59}{\sqrt{100}} = 1.66, \quad T = \frac{98.78 - 93.29}{1.66} \approx 3.32, \quad df = 99$$

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Two-sided p-value (via $2 * (1 - \text{pt}(3.32, df = 99)) \approx 0.00126$).

Since $p < 0.05$, we **reject** H_0 — convincing evidence that average run times genuinely changed between 2006 and 2017.

Wrapping up

- ▶ **Two-way tables** extend hypothesis testing beyond simple proportions to categorical variables with 3+ levels, using the chi-squared statistic — same randomization/mathematical-model logic, new test statistic and reference distribution.
- ▶ **A single mean** uses the exact same CLT/bootstrap/hypothesis-testing framework as a proportion — but swaps in the t-distribution wherever we'd have used the normal, because we're estimating σ with s .
- ▶ Next: this t-based approach extends to comparing **two** means (independent and paired) — the mean-based analog of Chapter 17.